

Kenan Blair

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EDUCATION

University of California, San Diego

B.S. in Artificial Intelligence, B.S. in Physics

La Jolla, CA

Sep. 2025 – Jun. 2029

North Hollywood High School

GPA: 4.6 SAT: 1560

North Hollywood, CA

Aug. 2021 – Jun. 2025

EXPERIENCE

Vertical Flight Society Software Lead

University of California, San Diego

October 2025 – Present

San Diego, CA

- Porting from NVIDIA Jetson Nano (Serial Comms, DDS) onboard computer to Pixhawk Flight Controller
- Designing precision landing algorithm (tf2, OpenCV, ROS2) and state machine (custom modes) for PX4 Autopilot
- Implementing competition simulation in Gazebo (3d modeling)
- **Leading team of 4:** managing multiple projects and timelines, fostering healthy work environment
- Built PX4 and ROS training modules for recruits

Triton Robotics Autonomy Member

University of California, San Diego

October 2025 – Present

La Jolla, CA

- Working with Kalman Filter predictions and temporal logic (C++) for ballistics
- Designed **Kalman filter** (FilterPy) to estimate enemy robot's angular position and velocity
- Learned ROS2 basics and associated robotics math (e.g. CMake, tf2, quaternions)

Chief Technology Officer

Hippocampi, Inc.

January 2025 – June 2025

Porter Ranch, CA

- Developed secure video telehealth platform on team of 2 devs for Chemobrain patients
- Raised **\$18,000**; spearheaded funding round for legal drafting and dev costs by pitching to angel investors
- Programmed **doctor recommendation system** w/ OpenAI API
- Programmed appointment system w/ T3 stack: payments, embedded notes, and MFA from scratch
- Oversaw projects and delegated technical tasks while compiling HIPAA documents w/ legal team

Neuroscience Research Assistant

University of California, Los Angeles

August 2024 – June 2025

Los Angeles, CA

- Engineered RL simulation of behavioral rat experiment **used by the research lab**, in Python from scratch
- Modeled tabular Q-learning algorithm to inform key insights on masking practices and state representations
- Communicated, catalogued, and published findings in 5,000 word paper

PROJECTS

Elements of Computing Systems | *HDL, Assembly, Compiler Logic, Systems Programming* June 2025 – January 2026

- Engineered fully functional processor (ALU, RAM, ROM) in HDL based on Von Neumann Architecture
- Built assembly language and respective 16-bit translator to interact w/ peripherals and registers
- Implemented stack logic with VM Translator (into assembly) to enable flow control and memory segments
- Designed compiler (Syntax Analyzer, Compiler Engine, etc) and programmed "Snake" with final product

YOLOV8 Object Detection from Scratch | *ML Theory, Real World Deployment*

February 2026 – Present

- Collected aerial training data and used SAM2 to preprocess
- Recreated YOLOV8 (CNN) architecture using torch tensors **without PyTorch Module class**
- Programmed and derived auto-diff and auto-grad algorithms (no library) to train the CNN
- **Deployed** object detection in our drone's computer vision stack for DBVF competition

TECHNICAL SKILLS

Languages: Java, C++, Python, C, Assembly, Javascript, HTML

Frameworks: React, Node.js, Flask, JUnit, Framer, NextJS, DrizzleDB, ROS2

Developer Tools: Linux/Unix CLI, Git, Docker, CMake, STM32IDE, Arduino

INTERESTS

Classical Piano, History, Surfing, Chess